S.No: 5	Exp. Name: ***Implement Sum of Subset Problem Using Backtracking***	Date:2026-08-31

ID: 25070521300

Aim:

You are given a set of integers and a target sum. Write a program that finds all subsets of the given set whose sum is equal to the target sum.

Input Format:

- The first line contains an integer, n the number of elements in the set.
- The second line contains n space-separated integers representing the elements of the set.
- The third line contains an integer, sum the target sum.

Output Format:

- Print all subsets whose sum is equal to the given target sum. Each subset should be printed in a new line, and the elements within the subset should be separated by a space.
- If no such subset exists, print "-1"
- The subsets should be printed in reverse order of their discovery (i.e., the last found subset should be printed first).

Example 1:

Input:

```
3
1 2 1
3
```

Output:

```
2 1
1 2
```

Example 2:

Input:

```
6
3 34 4 12 5 2
30
```

Output:

```
-1
```

Constraints:

- $1 \leq n \leq 20$
- $1 \leq element \leq 100$
- $1 \leq sum \leq 1000$

Source Code:

CTC64796.c

```
#include <stdio.h>

int arr[20], subset[20];
int result[1000][20], len[1000];
```

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```
int count = 0;

void findSubsets(int index, int n, int target, int sum, int k) {
    if (sum == target) {
        len[count] = k;
        for (int i = 0; i < k; i++)
            result[count][i] = subset[i];
        count++;
        return;
    }

    if (index == n || sum > target)
        return;

    subset[k] = arr[index];
    findSubsets(index + 1, n, target, sum + arr[index], k + 1);

    findSubsets(index + 1, n, target, sum, k);
}

int main() {
    int n, target;

    scanf("%d", &n);

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    scanf("%d", &target);

    findSubsets(0, n, target, 0, 0);

    if (count == 0) {
        printf("-1");
    } else {
        for (int i = count - 1; i >= 0; i--) {
            for (int j = 0; j < len[i]; j++)
                printf("%d ", result[i][j]);
            printf("\n");
        }
    }

    return 0;
}
```

Execution Results - All test cases have succeeded!

Test Case - 1
User Output
3
1 2 1
3
2 1
1 2

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Test Case - 2
User Output
6
3 34 4 12 5 2
30
-1